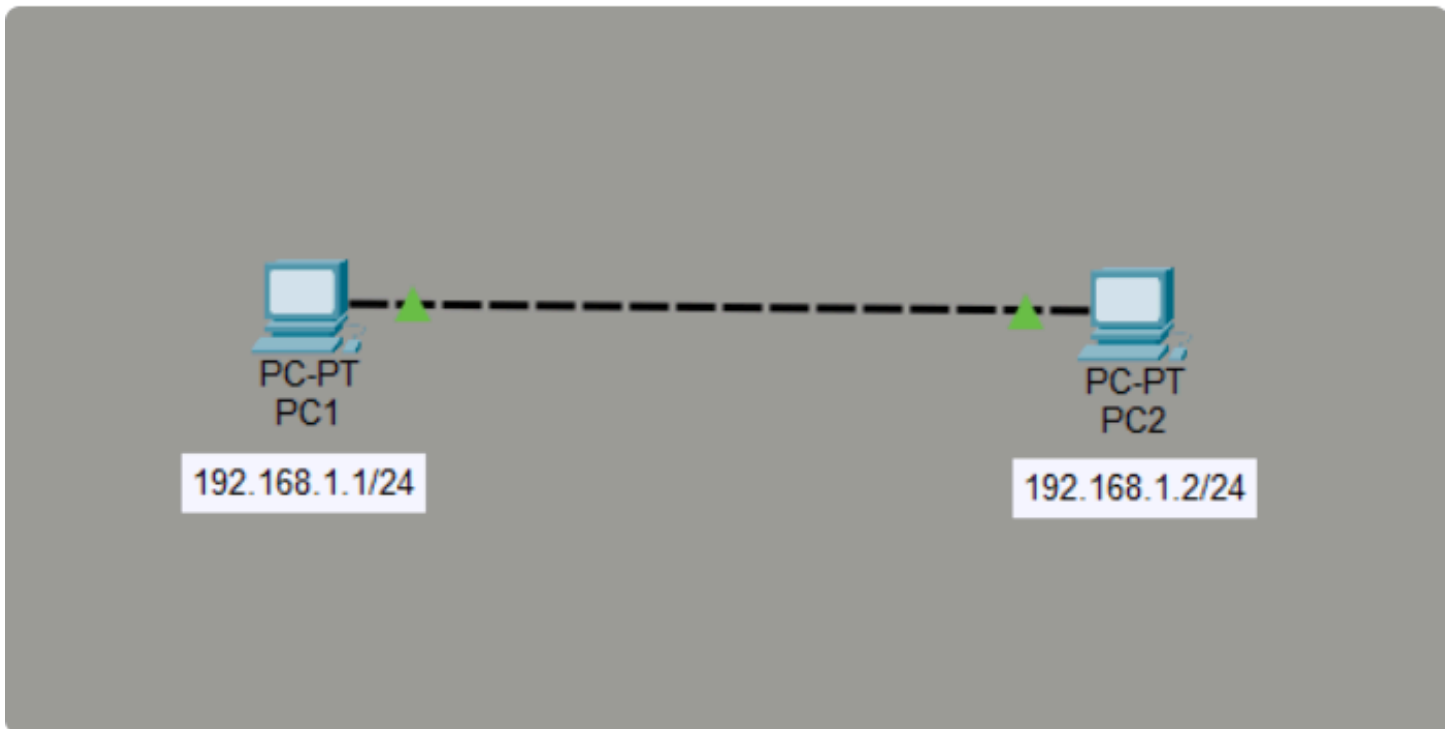


Assignment 3:

Objective:

- Design different network topologies
- Connect devices (PCs, switches, routers)
- Assign IP addresses
- Verify connectivity using **ping**

1. DIRECT PC TO PC:



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

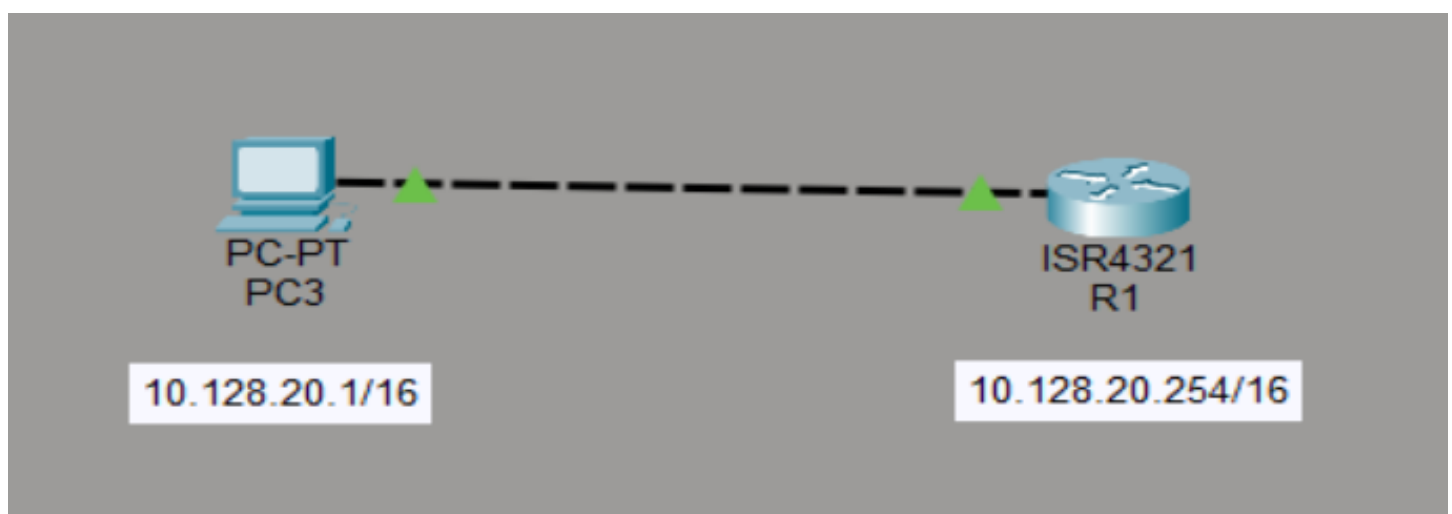
Pinging 192.168.1.2 with 32 bytes of data:

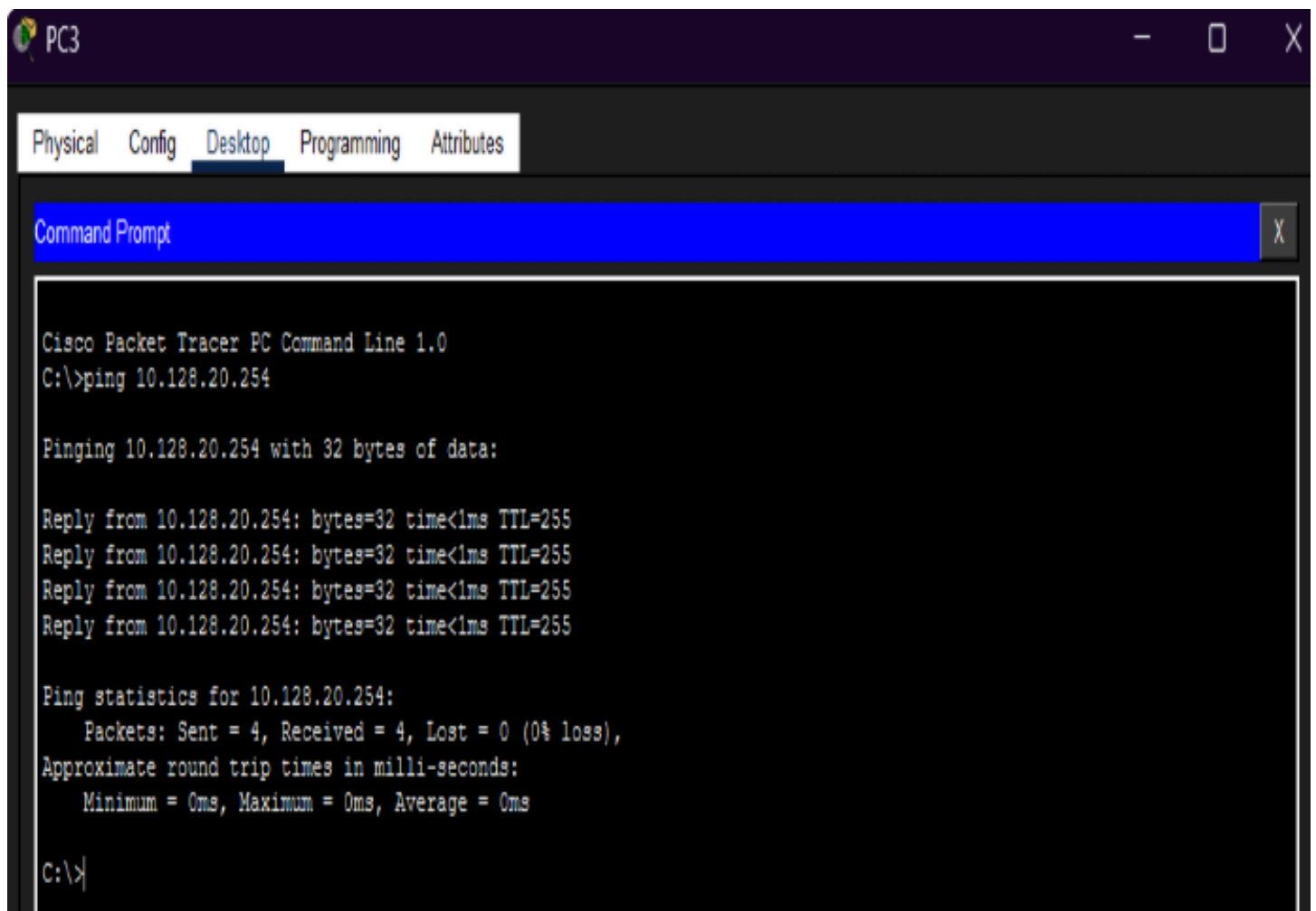
Reply from 192.168.1.2: bytes=32 time=8ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 8ms, Average = 4ms

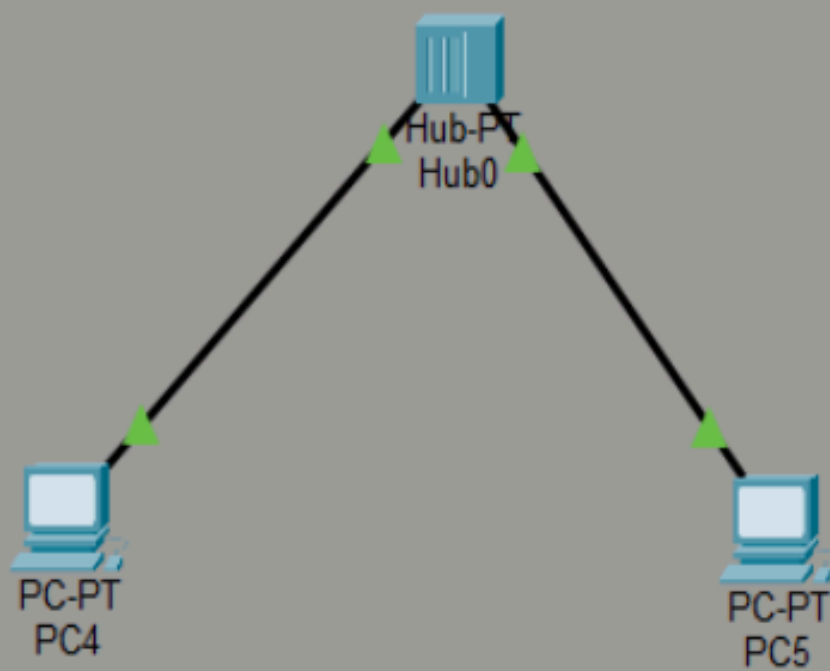
C:\>
```

2. PC TO ROUTER CONNECTION





3. TWO PCs using HUB



192.168.1.1/24

192.168.1.2/24

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=12ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128
Reply from 192.168.1.1: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 5ms

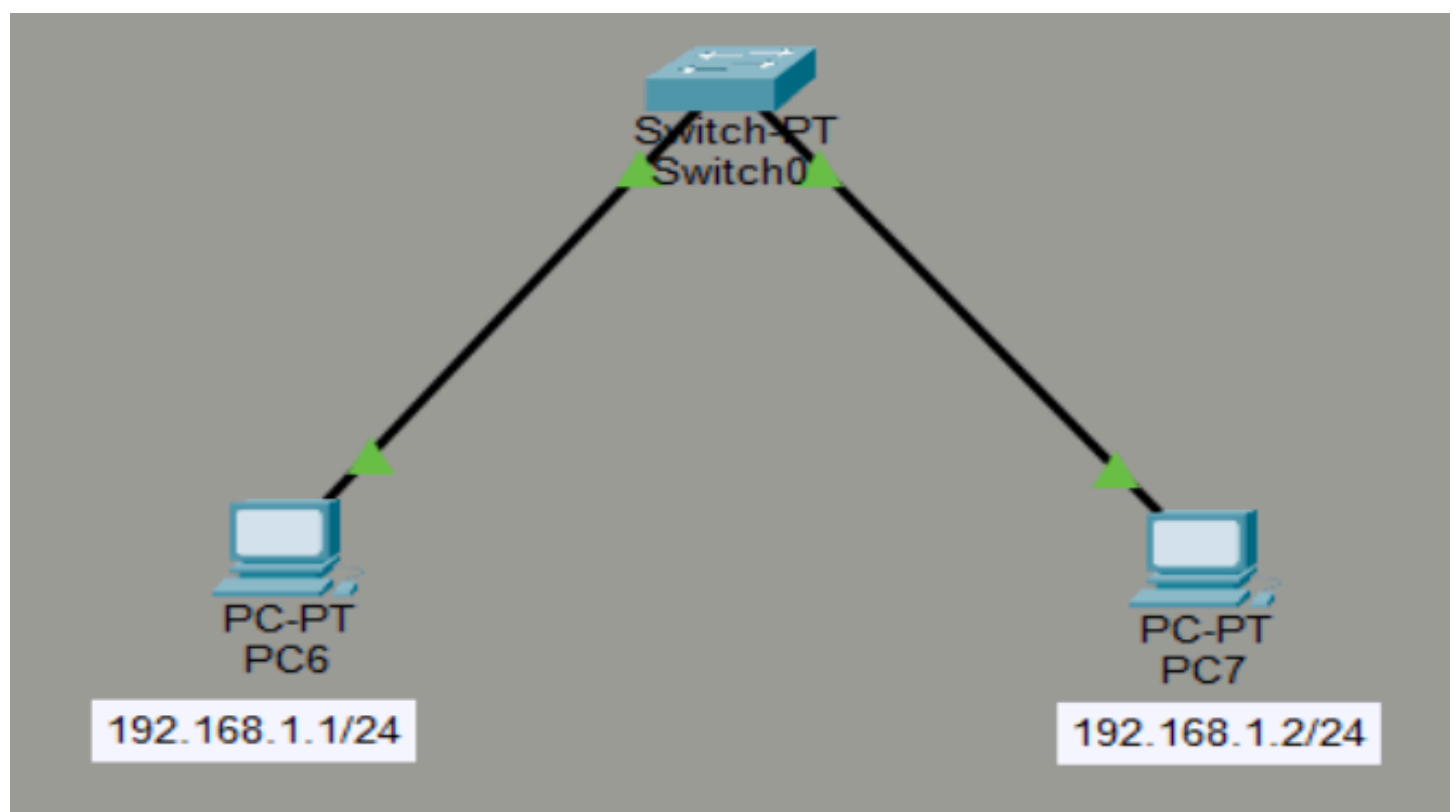
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. TWO PCs using switch



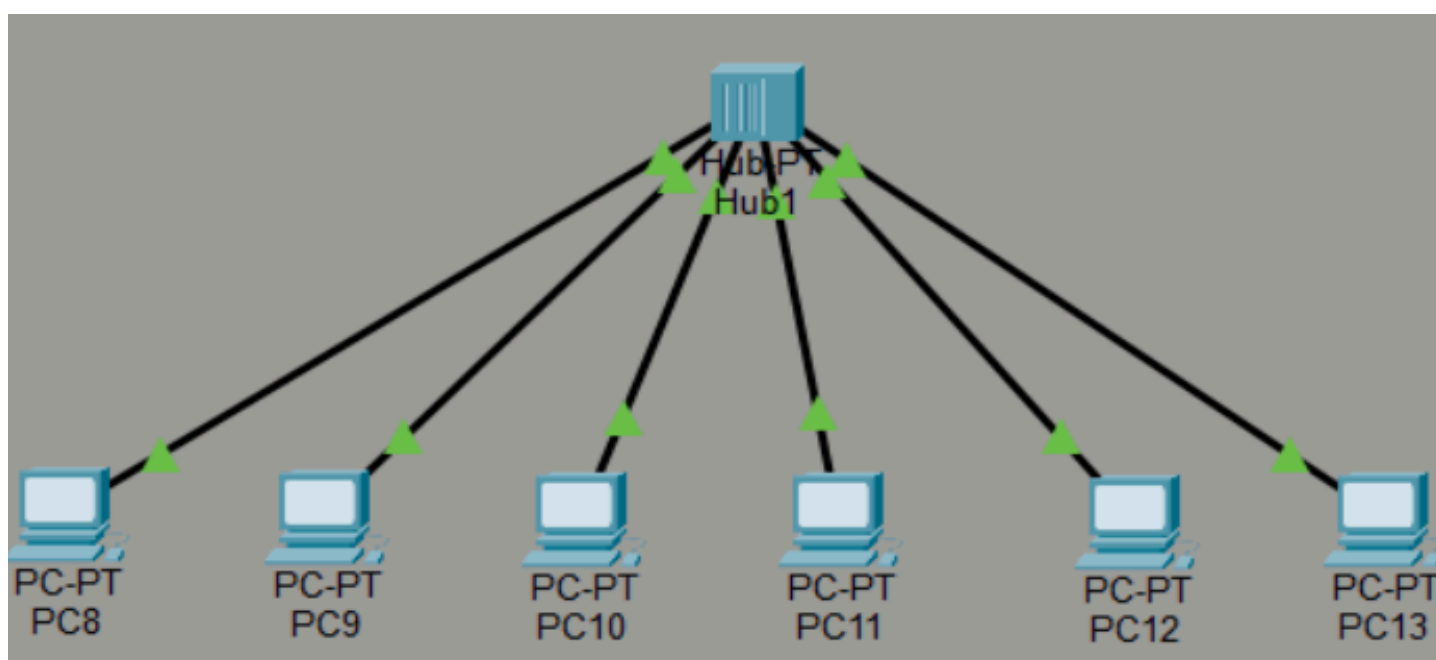
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5. BUS TOPOLOGY



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.1.5
```

```
Pinging 192.168.1.5 with 32 bytes of data:
```

```
Reply from 192.168.1.5: bytes=32 time=6ms TTL=128  
Reply from 192.168.1.5: bytes=32 time=6ms TTL=128  
Reply from 192.168.1.5: bytes=32 time=5ms TTL=128  
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.5:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 6ms, Average = 4ms
```

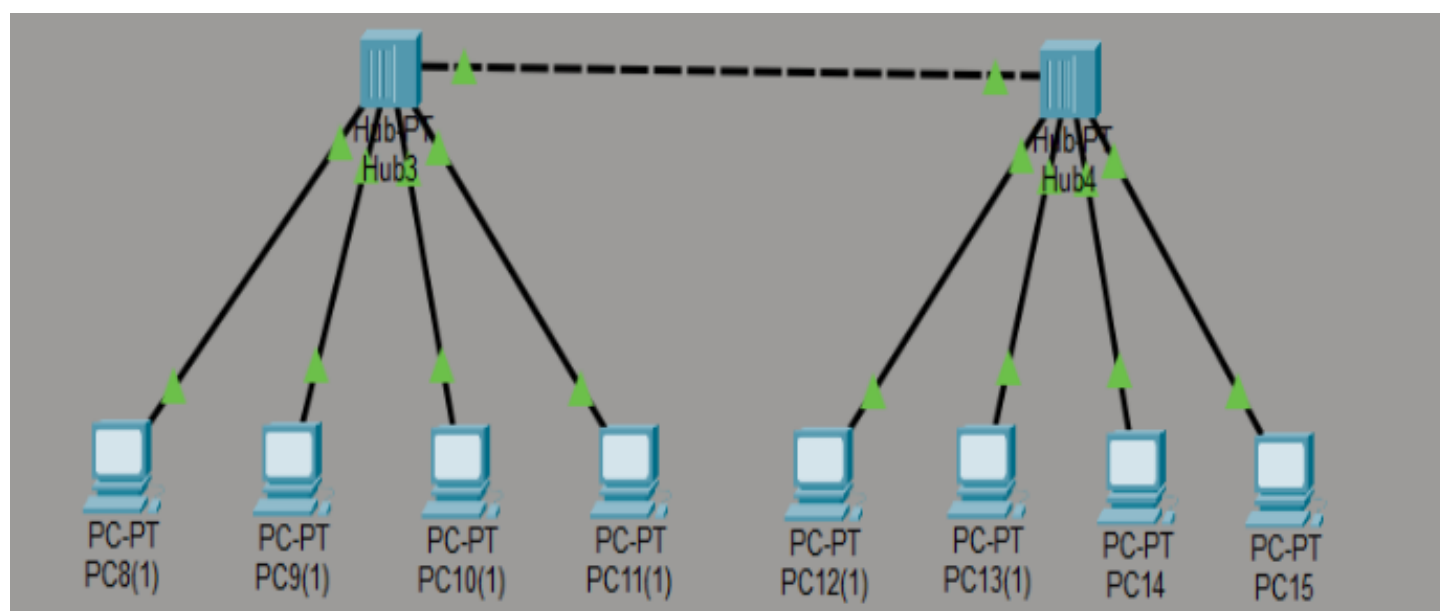
```
C:\>ping 192.168.1.7
```

```
Pinging 192.168.1.7 with 32 bytes of data:
```

```
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.7:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms


```
C:\>ping 192.168.1.8
```

```
Pinging 192.168.1.8 with 32 bytes of data:
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.8: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.1.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.1.5
```

```
Pinging 192.168.1.5 with 32 bytes of data:
```

```
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
```

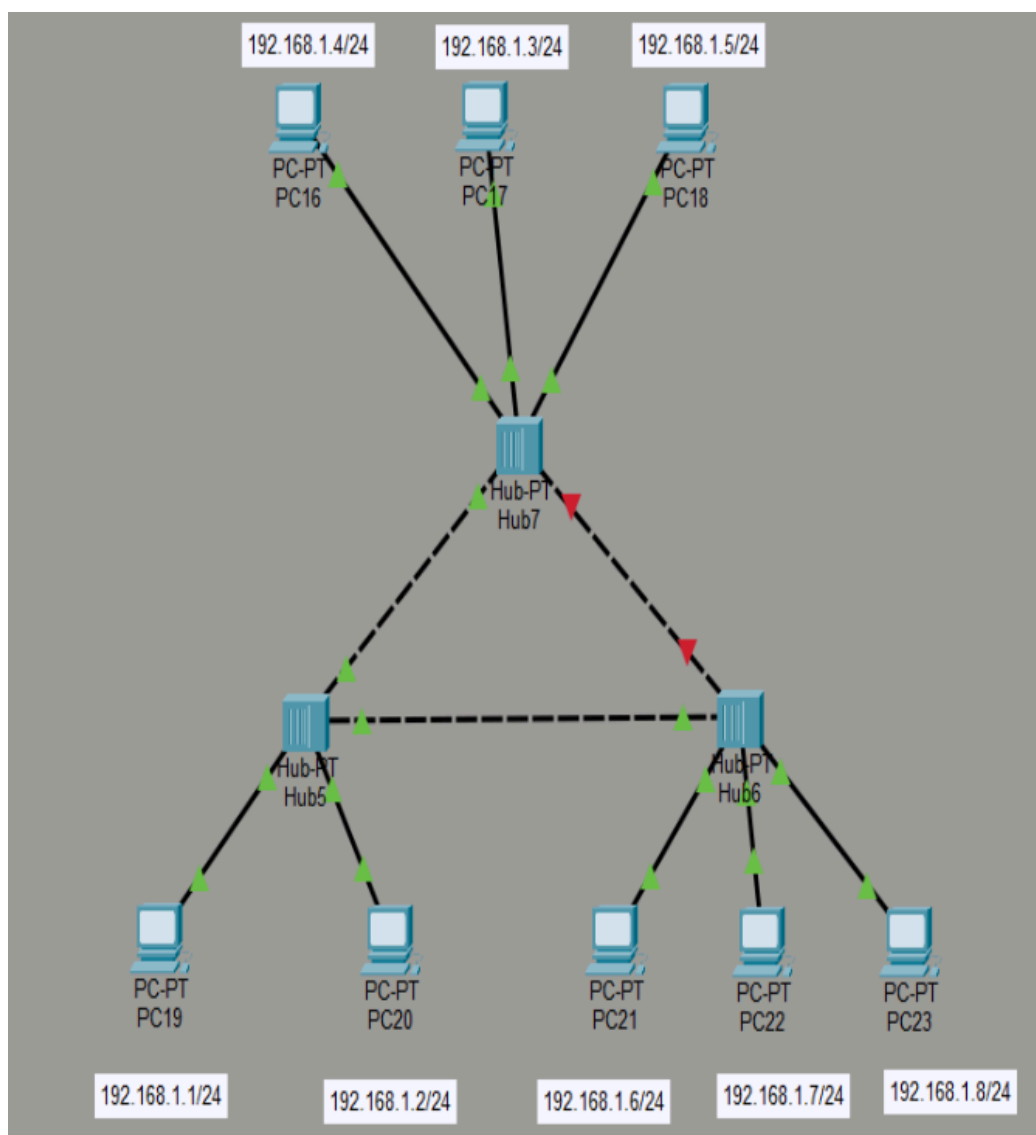
```
Ping statistics for 192.168.1.5:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

6. RING TOPOLOGY:



Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

```
C:\>ping 192.168.1.8
```

```
Pinging 192.168.1.8 with 32 bytes of data:
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

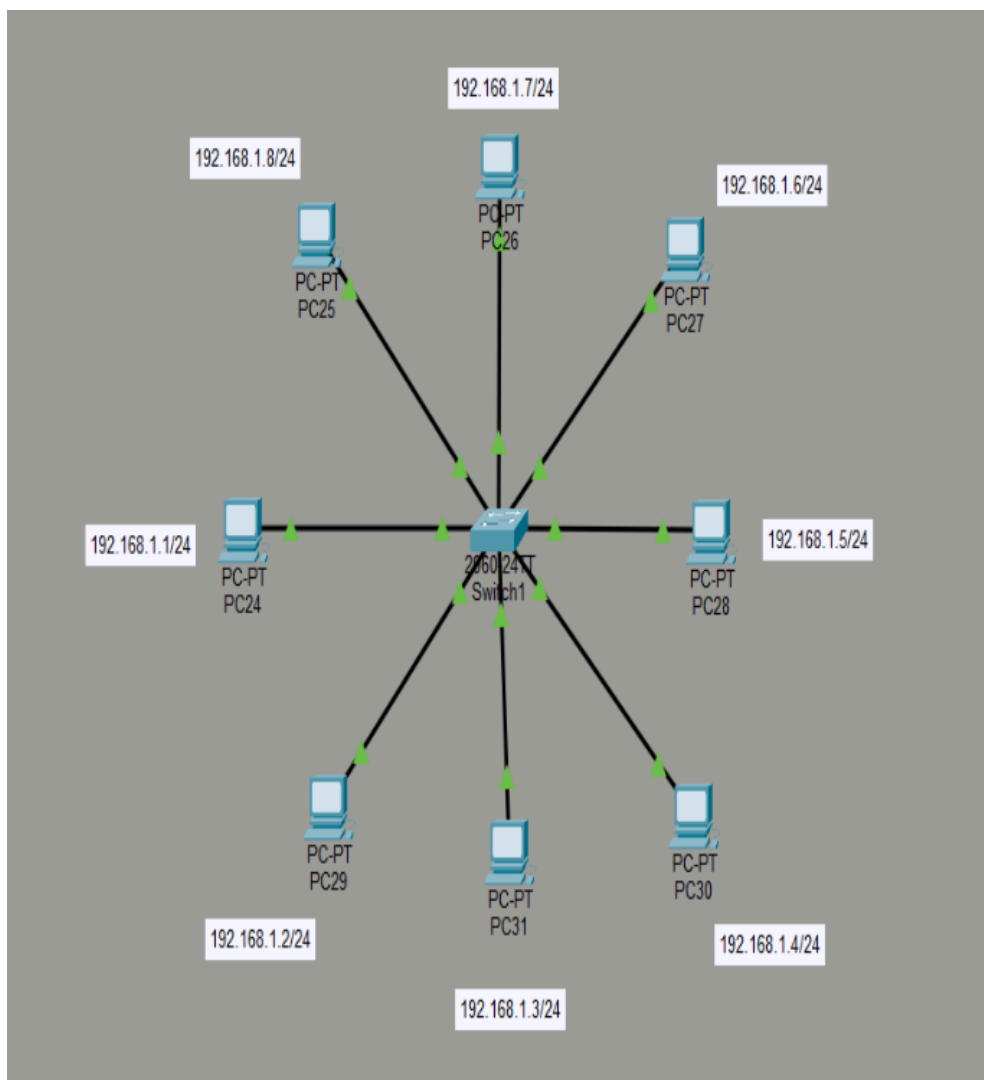
```
Ping statistics for 192.168.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

7. RING TOPOLOGY



Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=9ms TTL=128

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Reply from 192.168.1.4: bytes=32 time=5ms TTL=128

Reply from 192.168.1.4: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.4:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 9ms, Average = 4ms


```
C:\>ping 192.168.1.6
```

```
Pinging 192.168.1.6 with 32 bytes of data:
```

```
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.6: bytes=32 time=7ms TTL=128
```

```
Ping statistics for 192.168.1.6:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 7ms, Average = 1ms
```

```
C:\>ping 192.168.1.1
```

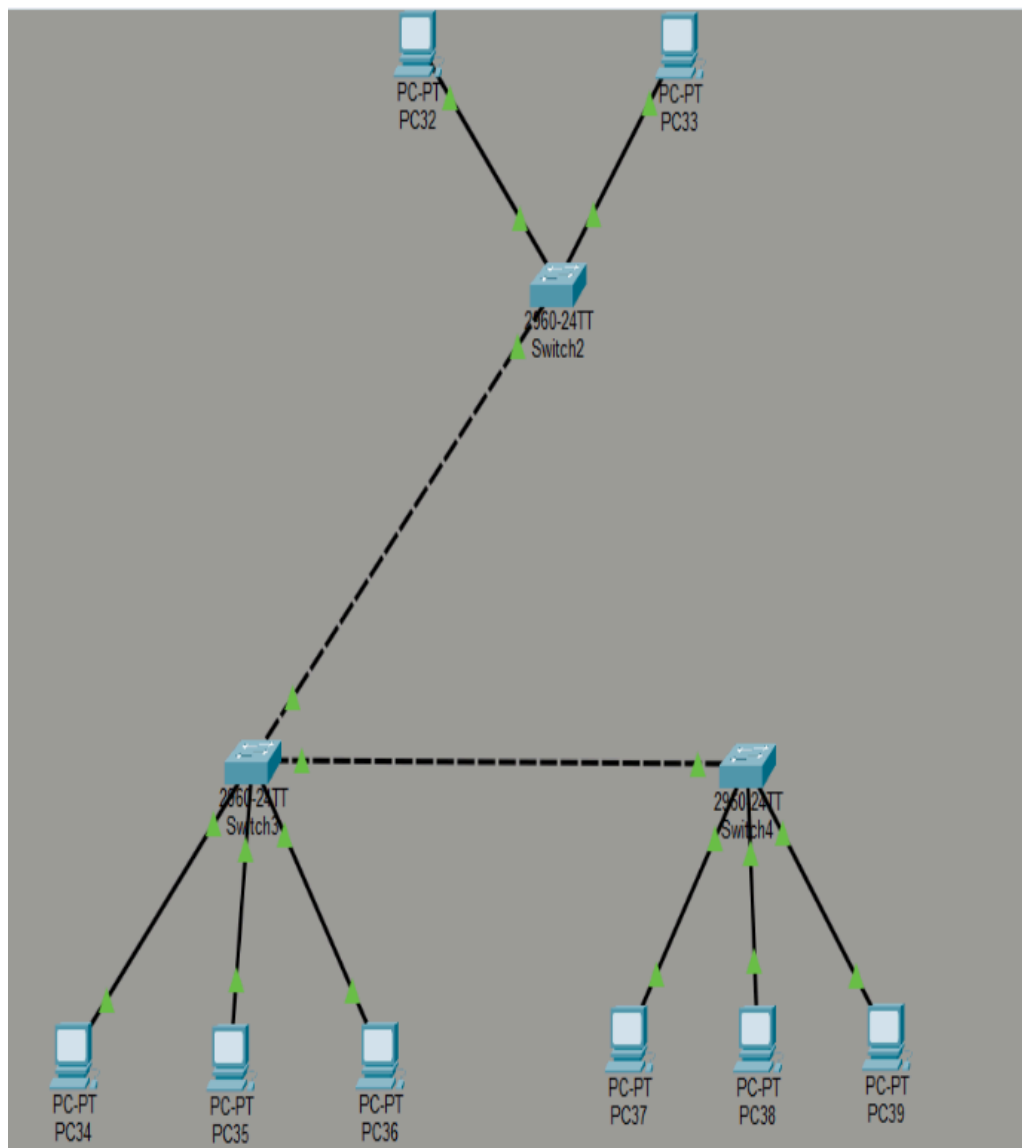
```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

8. TREE TOPOLOGY



Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Reply from 192.168.1.6: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.6:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 7ms, Average = 1ms

```
C:\>ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```